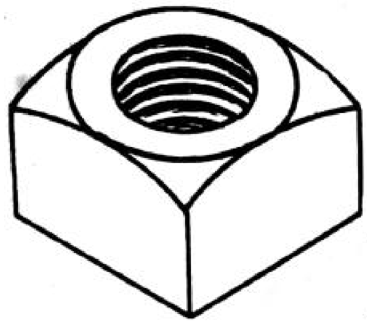
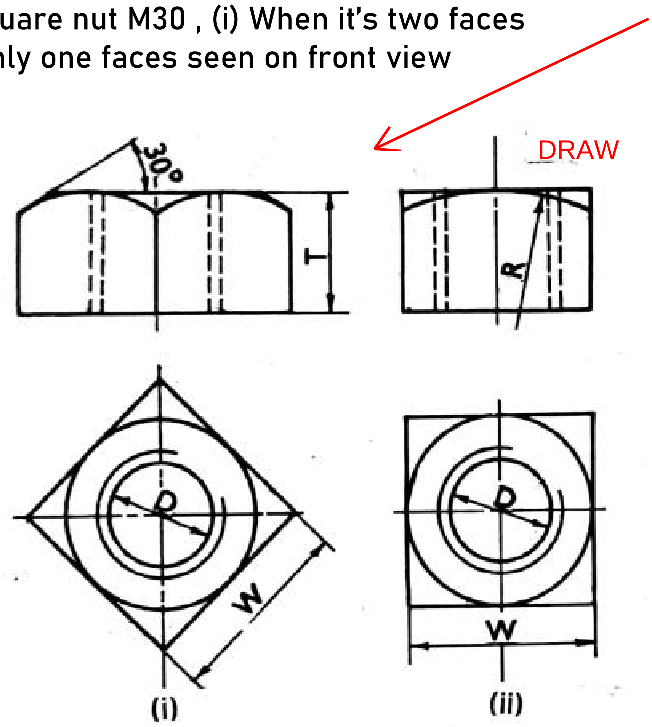
SQUARE NUT

1. Draw the front view and top view of a square nut M30 , (i) When it's two faces are equally seen in front view (ii) When only one faces seen on front view



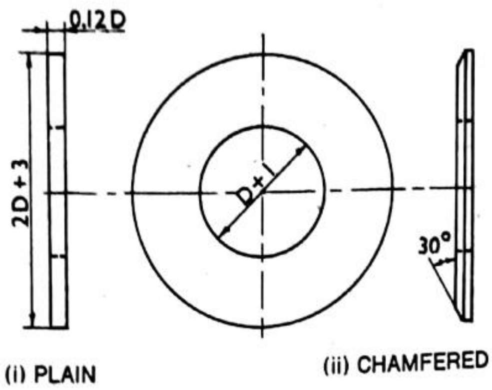


Washers

A washer is a cylindrical piece of metal placed below the nut to provide smooth bearing surface for the nut to turn on. It spreads the pressure of the nut over a greater area. It also prevents the nut from cutting into the metal and thus, allows the nut to be screwed-on more tightly. It is sometimes chamfered on the top flat surface.

TABLE 8-4
APPROXIMATE DIMENSIONS OF A WASHER

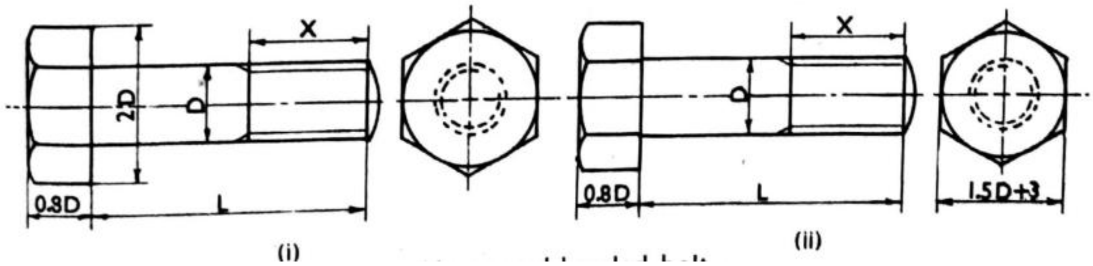
No.	Description	Dimensions
1.	Diameter of washer	$2D + 3$ mm
2.	Thickness	$0.12D$
3.	Angle of chamfer	30°
4.	Diameter of chamfer hole	$D + 0.5$ mm



Washers

Hexagonal - headed bolt

This is the most common form of a bolt. The hexagonal head is chamfered at its upper end. To prevent rotation of the bolt while screwing the nut on or off it, the bolt-head is held by another spanner.



2. Draw three views of a hexagonal-headed bolt, 24 mm diameter and 100 mm long, with a hexagonal nut and a washer.

Dimensions for the nut:

Thickness of the nut, *T* Ω 24 mm

Distance across diagonally opposite corners = 48 mm

Angle of chamfer = 30°

Radius of chamfer arc, *R* = 36 mm.

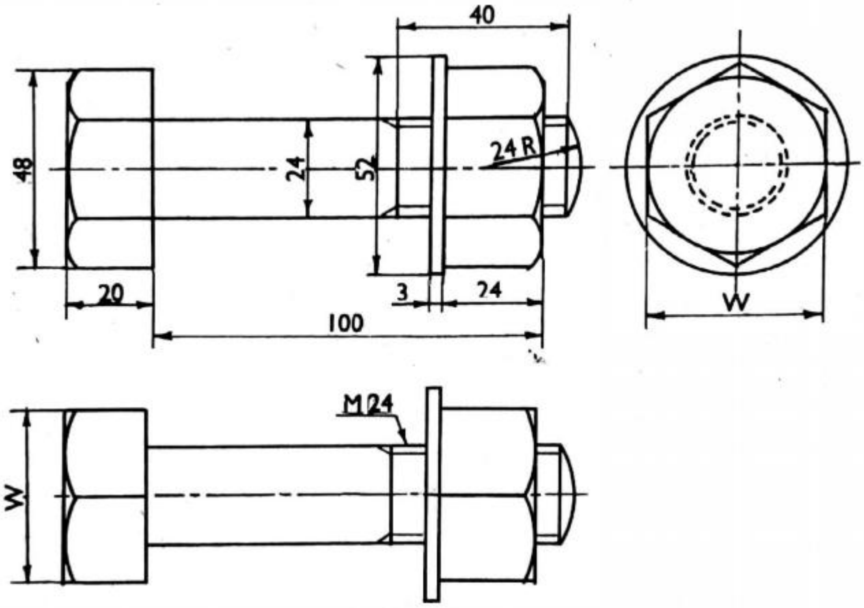
Dimensions for washer:

Diameter of the washer Ω 52 mm

Thickness Ω 3 mm

Diameter of the hole Ω 24.5 mm.

DRAW



Hexagonal-headed bolt with hexagonal nut and washer

Step 1: Draw the horizontal centre line and around it construct a rectangle 100 mm × 24 mm for the shank.

Step 2: Add the view of the bolt-head and the nut showing three faces. The distance between the outer edges will be 48 mm.

Step 3: Draw the chamfer arcs etc. as explained in problem 8-2.

Step 4: Draw the rectangle for the washer, 52 mm × 3 mm, attached to the nut.

Step 5: The end of the bolt is usually rounded. It is drawn by a radius equal to the diameter of the bolt.

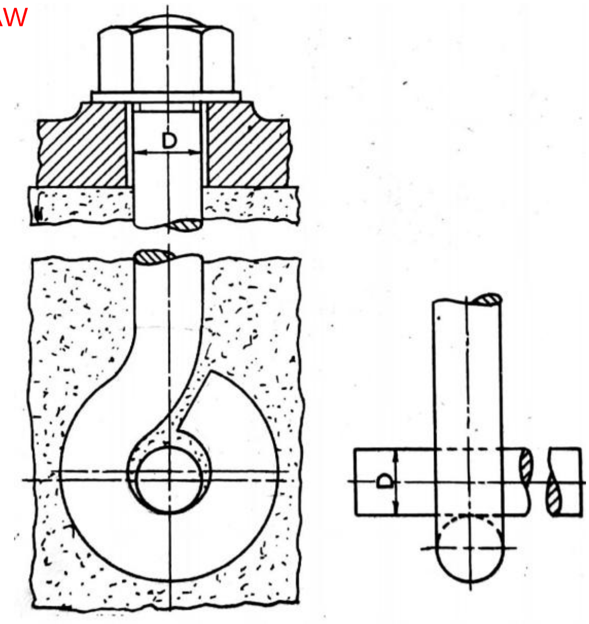
Step 6: Project the side view and the top view. The width *W* across the flats will be equal to $\sqrt{3} \times 24$ mm.

(iii) When drawing the views according to approximately standard dimensions, beginning must be made with the hexagon in the side view and the other views must then be projected from it. The distance *W* between the flat sides of the hexagon should be equal to $1.5D + 3$ mm.

Foundation bolts

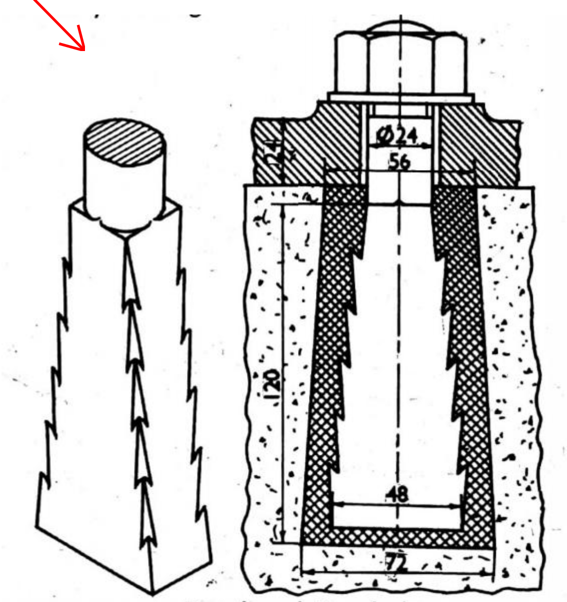
These bolts are used for fixing machines to their foundations.

DRAW



Eye foundation bolt

DRAW



Rag foundation bolt

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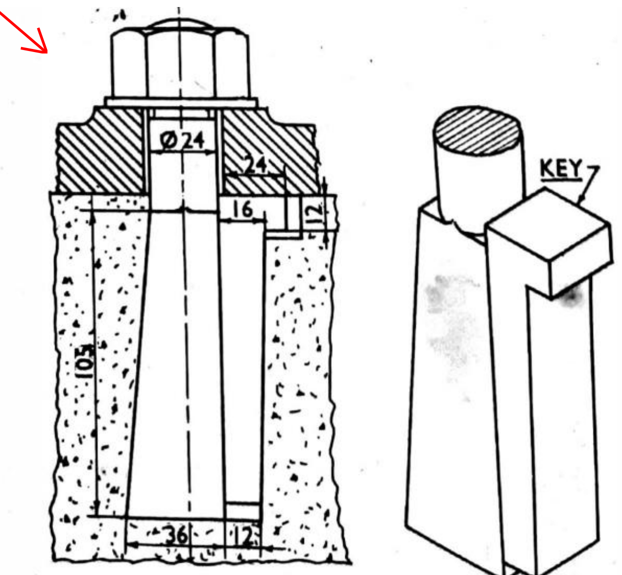
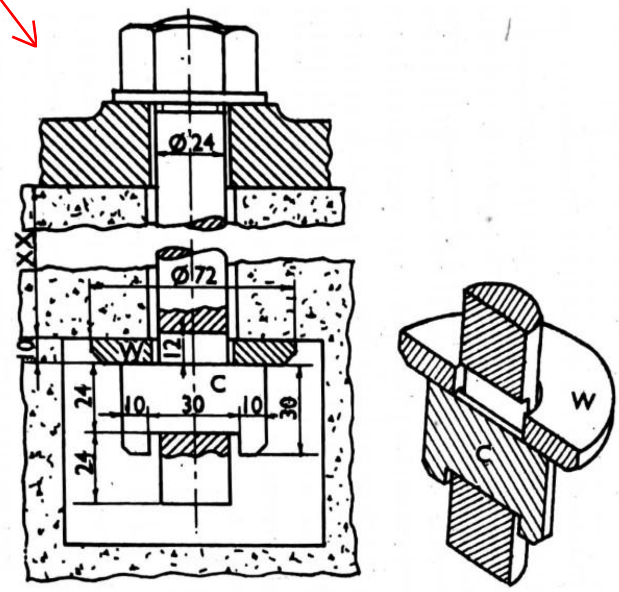


Lewis foundation bolt

DRAW



Cotter foundation bolt